PROJECT DOCUMENTATION

Artery Segmentation and Stenosis Detection using Advanced Deep Learning Techniques

1. Project Objectives

The primary objective of this project is to develop an automated, clinically reliable deep learning framework for accurate coronary artery segmentation and quantitative stenosis detection from medical angiographic images. Specifically, the system aims to:

- **Artery Segmentation:** Accurately delineate the lumen boundaries of coronary arteries in medical angiographic images, addressing challenges such as low contrast, vessel overlap, and background noise.
- **Centerline Extraction:** Extract continuous anatomical vessel centerlines to compute local cross-sectional diameters along the arterial tree.
- **Stenosis Detection & Localization:** Automatically detect, locate, and assess regions of arterial narrowing (stenosis) across major coronary branches.
- **Quantitative Severity Grading:** Calculate the percentage of diameter and area reduction (stenosis percentage) to determine lesion severity according to standard clinical criteria.

2. Workflow & Initial Preprocessing Pipeline

The project follows a structured sequential workflow from raw image acquisition to clinical diagnostic output:

- **Step 1: Data Preprocessing:** Image normalization, contrast enhancement (CLAHE), noise reduction, and region-of-interest (ROI) extraction from input angiographic scans.
- **Step 2: Vessel Segmentation:** Feeding preprocessed images into a deep learning segmentation model to generate binary/probability masks of the coronary arterial tree.
- **Step 3: Centerline & Diameter Profiling:** Morphological skeletonization of the predicted mask, curve smoothing, and orthogonal diameter calculation along the vessel length.
- **Step 4: Stenosis Detection & Quantification:** Identifying the minimal lumen diameter (MLD) relative to the reference healthy vessel diameter (RVD) to compute percentage stenosis: Stenosis % = (1 - MLD / RVD) * 100.
- **Step 5: Output & Reporting:** Visualizing segmented arteries, overlaying detected stenosis regions, and generating quantitative diagnostic metrics.

Initial Preprocessing Pipeline Details

- **Intensity Normalization:** Scaling raw pixel intensities to [0, 1] range and handling bit-depth variations (8-bit vs 12/16-bit DICOM).
- **Contrast Enhancement (CLAHE):** Contrast Limited Adaptive Histogram Equalization applied locally to boost low-contrast vessel contours without amplifying background noise.
- **Vesselness Filtering:** Multi-scale Frangi/Hessian filtering to enhance tubular vascular structures and suppress non-vascular planar artifacts.
- **Format Conversion & Resizing:** Standardizing multi-format inputs (DICOM, NIfTI, PNG, JPG) to uniform resolution (512x512) with isotropic spacing.

3. Imaging Modality, Dataset & Exploratory Data Analysis

- **Medical Imaging Modality:** Selected 2D X-ray Coronary Angiography (XCA) as the primary interventional modality for real-time cath-lab assessment, with complementary support for 3D Coronary Computed Tomography Angiography (CCTA).
- **Public / De-identified Datasets:** Selected and downloaded benchmark datasets: DCA1 (130 coronary angiograms with lumen masks), ARCADE (1,200 multi-view series with vessel tree and stenosis

annotations), and ASOCA (40 3D CCTA volumes).

- **Dataset Structure & Annotations:** Hierarchical directory structure containing raw images, expert-annotated binary lumen masks (0: background, 1/255: lumen), and stenosis metadata (bounding boxes, MLD, and % narrowing).
- **Exploratory Data Analysis (EDA):** Identified key data characteristics: severe foreground-to-background class imbalance (vessels occupy < 6% of pixels), non-uniform contrast agent distribution across cardiac phases, catheter occlusions, and patient respiratory motion artifacts.

4. Architecture & Full-Stack System Implementation

Deep Learning Model Architecture

- **Model Backbone:** Attention U-Net encoder-decoder network with multi-scale convolutional feature extraction layers.
- **Attention Mechanism:** Attention gates integrated into skip connections to prioritize vascular boundaries and suppress background tissue noise.
- **Loss Optimization:** Compound loss combining Focal Loss and Soft Dice Loss to resolve high foreground-background class imbalance.

Full-Stack System Architecture

- **Frontend:** React, TypeScript, and Vite providing a responsive, interactive clinical dashboard.
- **Backend:** FastAPI asynchronous REST API handling authentication, case lifecycle, upload handling, and analysis orchestration.
- **Database:** PostgreSQL storing users, cases, images, preprocessing runs, analyses, findings, and audit logs.
- **Storage:** Encrypted file storage for original uploads, processed tensors, segmentation masks, overlays, and model weights.
- **AI Services:** Decoupled inference services hosting separate models for 2D X-ray Angiography and 3D CT/CTA pipelines via a Model Registry.

Core Software Modules & Implementation

- **Authentication Module:** Secure user authentication using JWT sessions, password hashing (bcrypt), session expiration, and role-based access checks.
- **Case Management Module:** Comprehensive case lifecycle management (creation, patient code assignment, modality specification, status transitions, and team access).
- **Image Upload Module:** Secure file upload supporting drag-and-drop for multi-format medical imaging (DICOM, PNG, JPG, NIfTI, PDF) with validation and preprocessing triggers.

Database Schema

Table	Main Fields	Purpose
users	id, email, username, password_hash, role, status	User identity and role
user_profiles	user_id, display_name, organisation	Profile details
auth_sessions	id, user_id, expires_at, revoked_at	Remembered and active sessions
audit_events	id, user_id, action, metadata, created_at	Security and activity tracking
patients	id, patient_code, age_range, gender, risk_factors	De-identified patient details
cases	id, case_code, patient_id, modality, anatomy, status	Imaging case record
case_assignments	case_id, user_id, permission	Team access to a case
medical_images	id, case_id, source_path, format, spacing	Uploaded images and studies
preprocessing_runs	id, image_id, settings, output_path, status	Normalization & conversion history
analysis_runs	id, image_id, model_id, task, status, confidence	Each AI analysis execution
analysis_findings	id, analysis_id, type, location, severity, confidence	Vessel or stenosis findings
model_versions	id, name, version, modality, task, metrics, status	Registered models & validation metrics
reports	id, case_id, analysis_id, format, file_path	Generated PDF, JSON, CSV reports
notifications	id, user_id, case_id, level, message, read_at	Alerts for stenosis or failures

UI Wireframes & Role-Based Access Control (RBAC)

- **1. Login Screen:** Secure sign-in with email/password, social auth, and role switcher (Clinician, Researcher, ML Engineer).
- **2. Main Dashboard:** Navigation sidebar, summary counters (Total cases, In review, Alerts), and recent cases table.
- **3. Case & Upload Screen:** Form for Case ID, Modality (X-ray/CT/CTA), de-identified Patient Code, risk factors, drag-and-drop file upload, and preprocessing settings.
- **4. Analysis Workspace:** Interactive comparison viewer (Original image, Vessel mask, Stenosis overlay), centerline & diameter profile plots, confidence scores, and diagnostic findings.

Role	Main Access Permissions
Radiologist / Clinician	Review cases, view findings, add clinical comments, export reports
Medical Researcher	Upload de-identified data, compare analyses, export research data
ML Engineer	Dataset audit, preprocessing, training records, model registry management
Administrator	User & role management, audit logs, system settings, deleted-case recovery